Logistic Regression

Vidyadhar Upadhya

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- How to model $p(y = 1|x)$?

Logistic Regression-Statistical Perspective

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- The likelihood:

$$\begin{aligned} l(\theta|\mathcal{D}) &= p(Y|X;\theta) = \prod_{i=1}^{n} p(y^i|x^i;\theta) \\ &= \prod_{i=1}^{n} \left[h_\theta(x^i)\right]^{y^i} \; \left[1 - h_\theta(x^i)\right]^{1-y^i} \end{aligned}$$

Logistic Regression-Statistical Perspective

- The log-likelihood:

$$ll(\theta|\mathcal{D}) = \sum_{i=1}^{n} y^i \log h_\theta(x^i) + (1 - y^i) \log(1 - h_\theta(x^i))$$

- The maximum likelihood estimate of θ:

$$\hat{\theta} = \underset{\theta}{\mathit{argmax}} \quad ll(\theta|\mathcal{D})$$

- Iterative method can be used

Logistic Regression-Error Minimization Perspective

- We can also formulate logistic regression problem from the error minimization perspective using the cross entropy loss as below.
- Cross entropy loss:

$$L(y, \hat{y}) = \begin{cases} -\log \hat{y}, & \text{if } y = 1 \\ -\log(1 - \hat{y}), & \text{if } y = 0 \end{cases}$$

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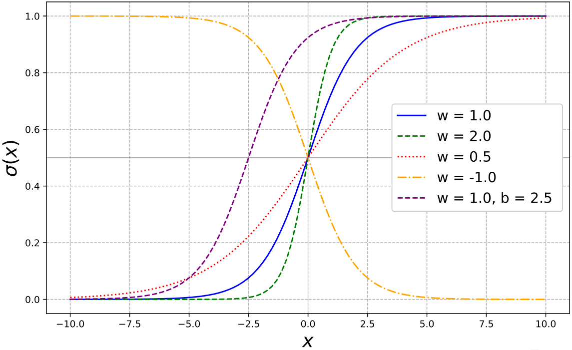
- More compactly, $L(y, \hat{y}) = -y \log \hat{y} - (1 - y) \log(1 - \hat{y})$
- Optimize loss over the training samples:

$$\hat{\theta} = \underset{\theta}{\mathit{argmin}} \quad -\sum_{i=1}^{n} \left[y^i \log \hat{y}^i + (1 - y^i) \log(1 - \hat{y}^i) \right]$$

Logistic Regression

- Note that $h_\theta(x)$ or $\hat{y}$ considered in previous slides can be any function whose range is $[0, 1]$
- One possible choice:

$$p(y = 1|x) = \sigma(x; w, b) = \frac{1}{1 + e^{-(wx+b)}}$$



Problem

- Assuming $h_\theta(x) = \sigma(\theta^T x)$, derive the update rule for MLE